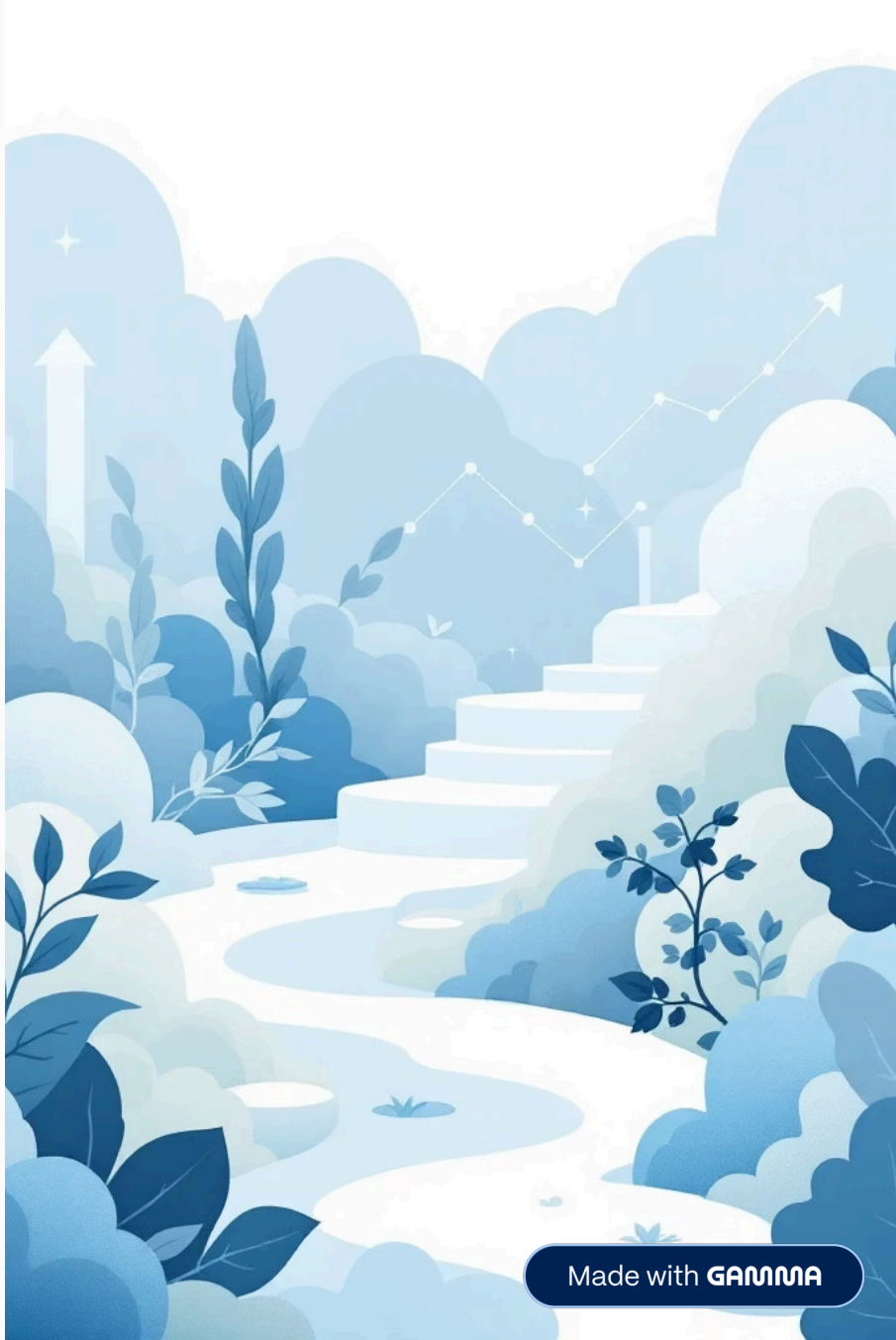
Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions.



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Project Overview

Analyze Data

Transactional data from 3,900 purchases.

Uncover Insights

Spending patterns, segments, preferences, subscriptions.

Guide Decisions

Inform strategic business choices.

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Dataset Summary

Data Points

- Rows: 3,900
- Columns: 18

Key Features

- Customer demographics
- Purchase details
- Shopping behavior

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Exploratory Data Analysis (Python)

Data preparation and cleaning in Python.

01

Data Loading

Imported dataset using `pandas`.

02

Initial Exploration

Checked structure and summary statistics.

03

Missing Data

Imputed `Review Rating` values.

04

Feature Engineering

Created `age_group` and `purchase_frequency_days`.

05

Database Integration

Loaded cleaned data to PostgreSQL.

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Data Analysis (SQL)

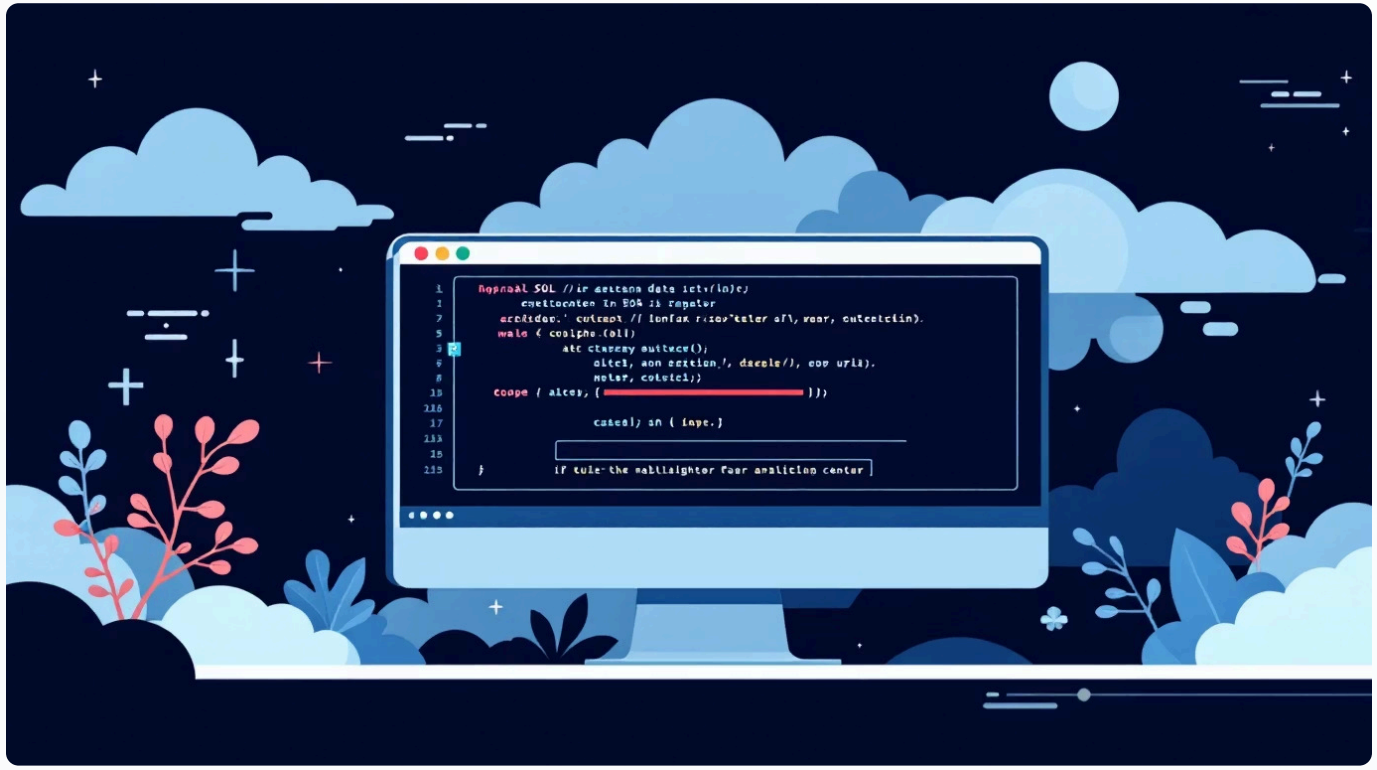
Key business questions answered with PostgreSQL.

Revenue by Gender

Female	75,191
Male	157,890

Top 5 Products by Rating

Gloves	3.86
Sandals	3.84
Boots	3.82



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SQL Insights: Shipping & Subscriptions

Shipping Type Comparison

Standard	58.46
Express	60.48

Subscribers vs. Non-Subscribers

Yes	1053 customers	59.49 avg spend
No	2847 customers	59.87 avg spend

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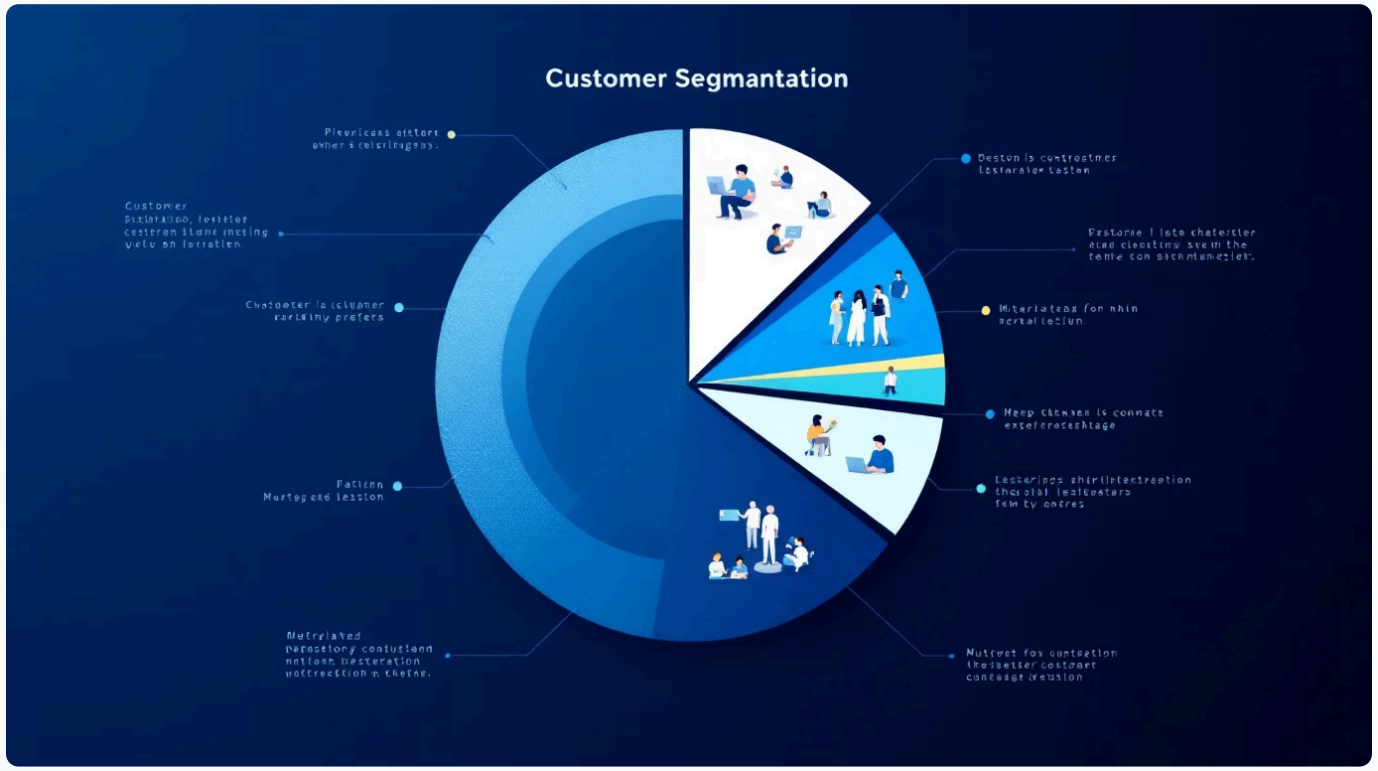
SQL Insights: Products & Segmentation

Discount-Dependent Products

Hat	50.00%
Sneakers	49.66%
Coat	49.07%

Customer Segmentation

Loyal	3116
New	83
Returning	701



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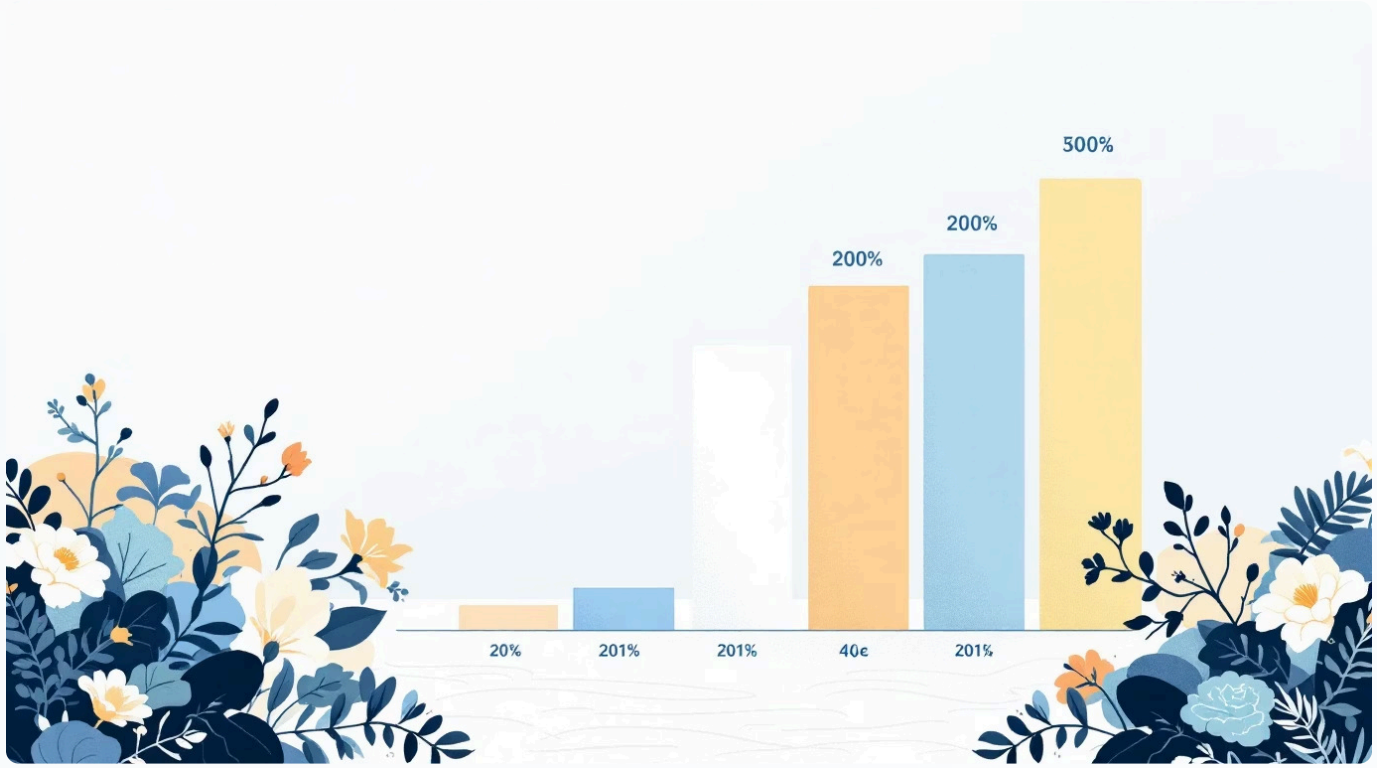
SQL Insights: Age & Repeat Buyers

Revenue by Age Group

Young Adult	62,143
Middle-aged	59,197
Adult	55,978

Repeat Buyers & Subscriptions

No	2518
Yes	958



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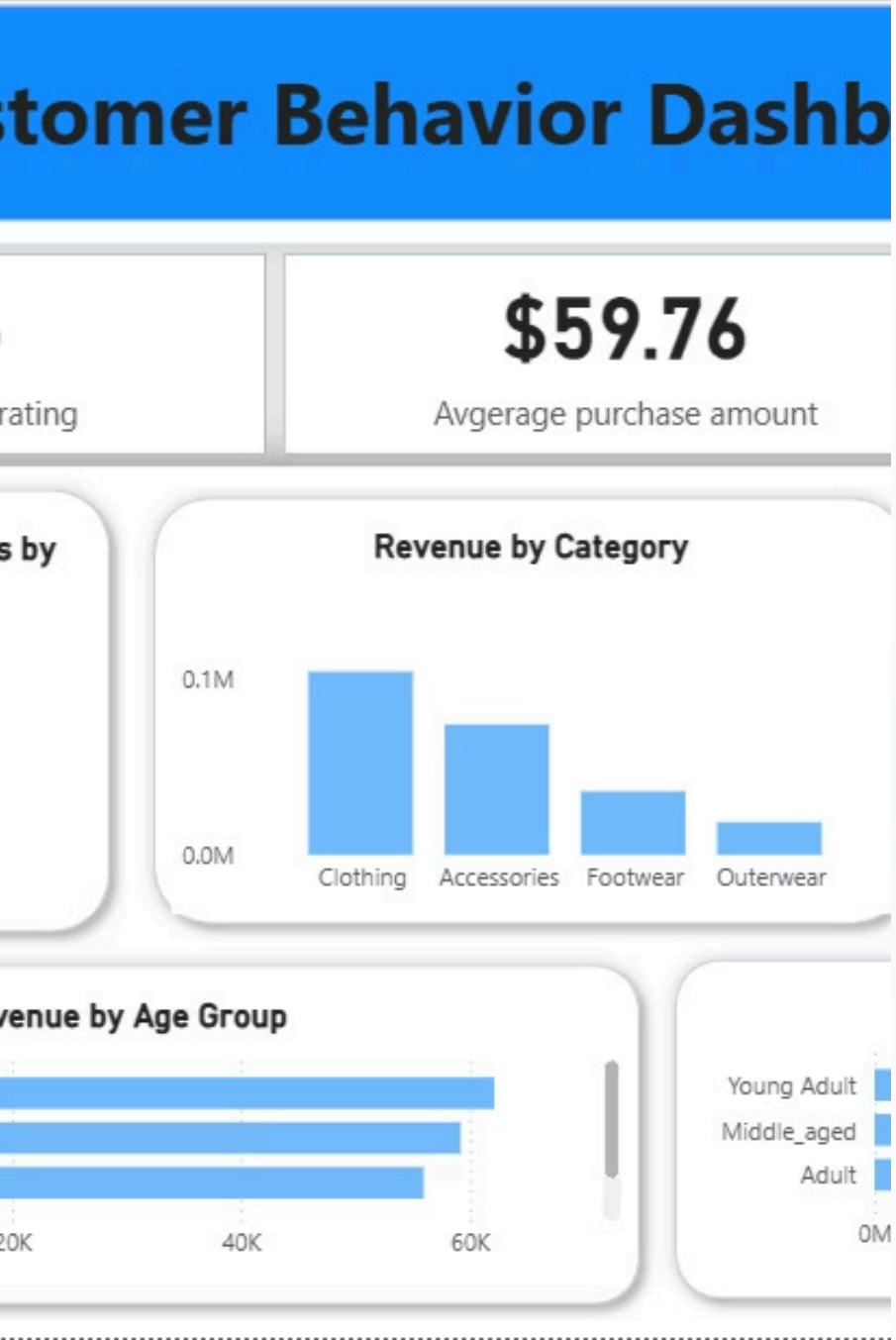

Power BI Dashboard

Interactive dashboard for visual insights.

3.9K	$59.76	3.75
Customers	Avg. Purchase	Avg. Review

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Business Recommendations

Boost Subscriptions

Promote exclusive benefits.

Customer Loyalty

Reward repeat buyers.

Review Discount Policy

Balance sales with margin.

Product Positioning

Highlight top-rated items.

Targeted Marketing

Focus on high-revenue groups.

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